Consumer Information and the Limits to Competition

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Roadmap

1. **Context** – Why information design matters for competition
2. **Model & Main Results** – Optimal signal structures for firms and consumers
3. **Discussion** – Extensions, limitations, and open questions

Motivation

Platforms design the information environment consumers face

- Amazon, Expedia, insurance comparison sites choose:
  - How much product detail to reveal
  - Whether to offer personalized recommendations
  - How flexibly consumers can filter and compare

- **Core question:** Should a platform (or regulator) make products *more* or *less* comparable?

- More transparency $\Rightarrow$ better product match, but higher prices
- Less transparency $\Rightarrow$ fierce price competition, but potential mismatch

Main Idea and Key Tension

Central trade-off: match quality vs. price competition

Firm-optimal policy

- Amplifies perceived differentiation
- Fewer price-sensitive consumers
- Higher equilibrium prices
- No mismatch $\Rightarrow$ max total welfare

Consumer-optimal policy

- Dampens perceived differentiation
- More price-sensitive consumers
- Lower equilibrium prices
- Some mismatch $\Rightarrow$ welfare loss

Contribution to the Literature

Information design meets oligopoly pricing

- **Monopoly information design** (Roesler & Szentes, 2017)
  → Extends to duopoly; competition changes the problem fundamentally
- **Bayesian persuasion** (Kamenica & Gentzkow, 2011)
  → Applies persuasion framework to competitive pricing
- **Competitive disclosure** (Anderson & Renault, 2006; Johnson & Myatt, 2006)
  → Goes beyond "truth or noise" to general signal structures
- **Centralized vs. decentralized design** (Ivanov, 2013; Hwang et al., 2019)
  → Centralized design allows *relative* signals (e.g., "top product")

Model Setup

- Two firms ($i = 1, 2$) supply differentiated products at zero cost
- Consumer values product i at $v_i \geq 0$; prior on (v_1, v_2) is symmetric
- Only the **relative preference** matters: $x \equiv v_1 - v_2$
  - Prior CDF $F(x)$ symmetric around 0, support $[-1, 1]$
- Consumer observes a **private signal** s before purchase
  - Signal structure $\{\sigma(s|x), S\}$ is public and chosen *before* firms set prices
- After signal s, consumer updates to **posterior** $G(x) \equiv$ CDF of $E_{F_s}[x]$
- Firms observe G but not s; set prices simultaneously (Bertrand–Nash)
- Consumer buys from firm 1 if $E[x \mid s] > p_1 - p_2$

Key Constraint: Mean-Preserving Contraction

Bayesian consistency: any posterior G must be a **mean-preserving contraction** (MPC) of the prior F:

$$\int_{-1}^{x} G(\tilde{x})\, d\tilde{x} \;\leq\; \int_{-1}^{x} F(\tilde{x})\, d\tilde{x} \qquad \text{for all } x \in [-1, 1]$$

- Signals can reshuffle beliefs inward, but cannot create information from nothing
- Any G satisfying MPC can be induced by some signal structure
- $\Rightarrow$ Instead of optimizing over signals, we optimize directly over the posterior G

Bounds on Equilibrium Posteriors

Lemma 1

G implements symmetric price p in pure-strategy equilibrium **iff**

$$L_p(x) \;\leq\; G(x) \;\leq\; U_p(x) \qquad \forall\, x \in [-1, 1]$$

Upper bound (from firm 2's no-deviation condition):

- Firm 2 deviates to p_2'. Profit from deviation: $p_2' \cdot G(p_1 - p_2')$
- No deviation is profitable iff $p_2'\, G(p_1 - p_2') \leq \pi_2 = p/2$ for all p_2'
- Substituting $x = p - p_2'$ and rearranging:

$$G(x) \;\leq\; \frac{p}{2(p-x)} \;\equiv\; U_p(x) \qquad \text{for } x < \tfrac{1}{2}p$$

Lower bound $L_p(x) = 1 - \frac{p}{2(p+x)}$: symmetric argument for firm 1

L_p and U_p are mirror images: $L_p(x) = 1 - U_p(-x)$

Symmetric Signals Dominate

Lemma 2

Relative to any *asymmetric* posterior (one that treats firms differently):

(i) There exists a **symmetric** posterior with **higher profit for each firm**

(ii) There exists a **symmetric** posterior with **higher consumer surplus**

Intuition:

- Biased signals favor one firm $\Rightarrow$ the other undercuts aggressively
- The resulting price war hurts *both* firms — even the favored one
- For consumers: asymmetry causes mismatch + price dispersion

$\Rightarrow$ **We can restrict attention to symmetric signal structures**

Firm-Optimal Signal: Intuition

Goal: find the highest sustainable price p

1. Higher $p \Rightarrow$ the feasibility corridor $[L_p, U_p]$ becomes flatter
2. To fit inside a flat corridor *and* satisfy MPC, G is squeezed against the **lower bound** L_p on $[-1, 0]$
3. $G = L_p \Rightarrow$ **few near-indifferent consumers** (U-shaped density)
   - Neither firm can profitably undercut $\Rightarrow$ high price is sustainable
   - Everyone has strong preferences $\Rightarrow$ buys preferred product $\Rightarrow$ **no mismatch**

Push p until MPC binds $\Rightarrow$ **the maximum feasible price**

Firm-Optimal Signal: Result

Proposition 1

The firm-optimal signal induces the symmetric equilibrium price:

$$p^* = \frac{2\delta}{1 - \log 2} \approx 6.5\,\delta$$

where $\delta \equiv E_F[\max\{x, 0\}]$ measures how differentiated products are.
Implemented by $G(x) = L_{p^*}(x)$ for $x < 0$.

- Higher δ (more differentiation) $\Rightarrow$ higher optimal price
- **No mismatch** $\Rightarrow$ total welfare is maximized $(= \mu + \delta)$
- Profit $\geq 63\%$ higher than under full disclosure

Firm-Optimal Posterior (Uniform Prior)

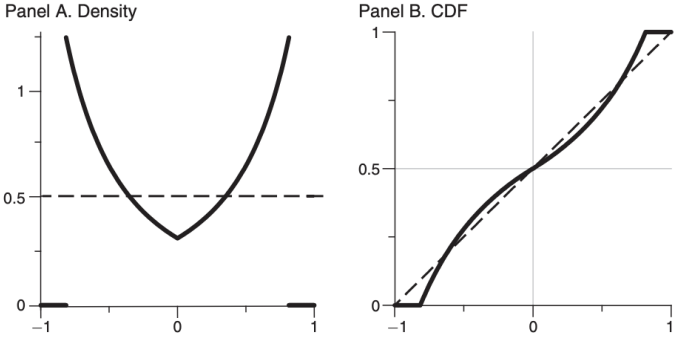

FIGURE 5. FIRM-OPTIMAL POSTERIOR

Dashed = uniform prior. Solid = firm-optimal posterior $G = L_{p^*}$.

U-shaped density: mass at extremes, few indifferent consumers $\Rightarrow$ high price.

Consumer-Optimal Signal: Intuition

Goal: maximize consumer surplus = match value − price

1. Consumers want fierce price competition $\Rightarrow$ need **many near-indifferent consumers**
2. This means pushing G against the **upper bound** U_p on $[-1, 0]$
   - Density peaked near $x = 0 \Rightarrow$ firms terrified of losing swing consumers
   - Each firm undercuts aggressively $\Rightarrow$ very low price
3. **Cost:** some near-indifferent consumers buy the wrong product (mismatch)
4. But the **price drop more than compensates** for match quality loss

Consumer-Optimal Signal: Result

Proposition 2

The consumer-optimal signal induces price:

$$p^{**} = \frac{-\gamma}{1-\gamma} F^{-1}\left(\frac{\gamma}{2}\right), \qquad \gamma \approx 0.05$$

Implemented by: $G(x) = \min\left\{F(x),\ U_{p^{**}}(x)\right\}$ for $x < 0$

- **Mismatch occurs** — total welfare $< \mu + \delta$
- Price effect dominates $\Rightarrow$ higher consumer surplus than full disclosure
- $G = \min\{F, U_p\}$: posterior follows the prior until it hits the upper bound, then rides along it — blurring only weak preferences

Consumer-Optimal Posterior (Uniform Prior)

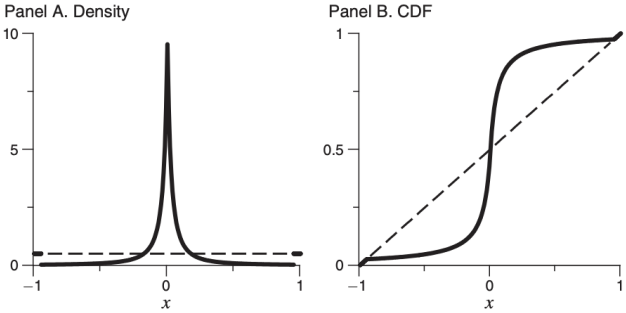

FIGURE 8. CONSUMER-OPTIMAL POSTERIOR

Dashed = uniform prior. Solid = consumer-optimal posterior.

Spike near $x = 0$: many nearly indifferent consumers $\Rightarrow$ very low price.

The Limits to Competition

Proposition 3

Any profit $p \in [0, p^*]$ is feasible. For a given profit level p:

- **Min CS:** $\mu - \frac{1}{2}(1 + \log 2)\, p \quad$ (posterior $= L_p$)
- **Max CS:** $\mu + \delta - p$ when $p \geq p_F$; concave frontier when $p < p_F$

Key implication: competition makes it **impossible** to have both low profit *and* low CS simultaneously

- Full disclosure optimal *only* for unweighted total welfare
- Ramsey problem picks a point on the feasible frontier
- Contrast with monopoly (Roesler & Szentes, 2017): there low profit *and* low CS is feasible

Feasible Combinations of Profit and Consumer Surplus

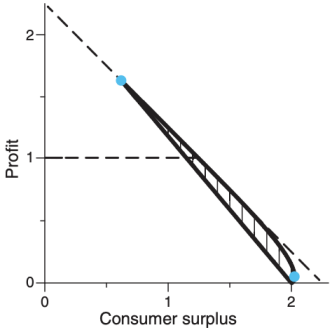

FIGURE 9. FEASIBLE COMBINATIONS OF PROFIT AND CONSUMER SURPLUS

Shaded area = feasible (profit, CS) pairs. Upper-left dot = firm-optimal; lower-right dot = consumer-optimal. Dashed line = max total welfare $\mu + \delta$.

Due to competition, the feasible set cannot reach the origin.

Summary of Results

	Firm-optimal	**Consumer-optimal**
Price	$p^* = \frac{2\delta}{1-\log 2} \approx 6.5\delta$	$p^{**} \approx 0.05\, p_F$
Posterior	$G = L_{p^*}$ (lower bound)	$G = \min\{F, U_{p^{**}}\}$
Mismatch?	**No**	**Yes**
Total welfare	Maximized $(\mu + \delta)$	Not maximized
Symmetric?	Yes	Yes

- More information $\neq$ more competition
- Biased recommendations never help either side (Lemma 2)
- Competition prevents simultaneously low profit *and* low CS (Prop. 3)

Extensions

1. **Mixed-strategy equilibria**
   - Allowing mixed strategies improves CS by at most 1.6%
   - Pure-strategy restriction is essentially without loss

2. **More than two firms**
   - "Top product" signal: can achieve first-best profit for firms
   - "Top two" signal: induces Bertrand $\Rightarrow p = 0$, best for consumers
   - As $n \to \infty$: trade-off between price and match quality vanishes

Discussion and Open Questions

- Scalar heterogeneity ($x = v_1 - v_2$): multi-dimensional preferences are harder
- Symmetric firms: what if one firm is known to be better (vertical differentiation)?
- Information designer as strategic agent (platform with own objective)?
- Endogenous product design: firms may adjust products in response to signals
- **Find extensions that are easily implemented.**